Grammatical Metaphor Increases AI-Animacy Perception in Online Sentence Processing

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BACKGROUND

- Subject position of transitive clauses carries proto-agentive entailments, such as causativity and animacy.[1,2]
- English allows this agentive subject position to be occupied by non-animates in a phenomenon known as **grammatical metaphor (GM)**.[3]
- **GM** increases the responsibility assigned to non-animates in rating tasks.[4,5,6]
- It is not known whether **GM** relies on anthropomorphic beliefs.

RESEARCH QUESTIONS

1. How does **GM** affect online sentence processing?
2. Does **GM** increase perception of AI animacy?
3. How does the effect of **GM** interact with anthropomorphic beliefs?

HYPOTHESIS

Grammatical metaphor will increase acceptance of AIs as subjects of animate verbs for participants with anthropomorphic beliefs.

EXPERIMENT DESIGN

Independent Variables
- Framing Condition: **GM** / **no GM**
- Explanation Verb: **Animate** / **Inanimate**
- Technology Type: LLM / other AI / non-AI
- Outcome: Positive / Negative

Participant Variable:
- Technology-Individual Differences in Anthropomorphism Questionnaire (**t-IDAQ**) (Modified version of the IDAQ,[7] focused on AIs, LLMS, and Computer Programs)

STIMULI

36 Critical Items, 24 fillers (Ex. 2 Only)

GM	no GM
A text processing A.I. called Lo-Go™ massively improved Quantum Mutual Funds popularity after they launched it. Lo-Go™ generated summaries of thousands of earnings calls… In the first three months, Lo-Go™ lost billions for investors.	By launching a text processing A.I. called Lo-Go™, Quantum Mutual Funds massively improved their popularity. By using Lo-Go™, Quantum Mutual Funds generated summaries of thousands of earnings calls… With Lo-Go™, investors lost billions in the first three months.
Animate Explanation	**Inanimate Explanation**
It all happened because Lo-Go™ had [judged] the earnings calls incorrectly.	It all happened because Lo-Go™ had [decoded] the earnings calls incorrectly.

[ ] marks the critical region for eye-tracking measures

METHODS & PARTICIPANTS

Experiment 1 (Ratings)

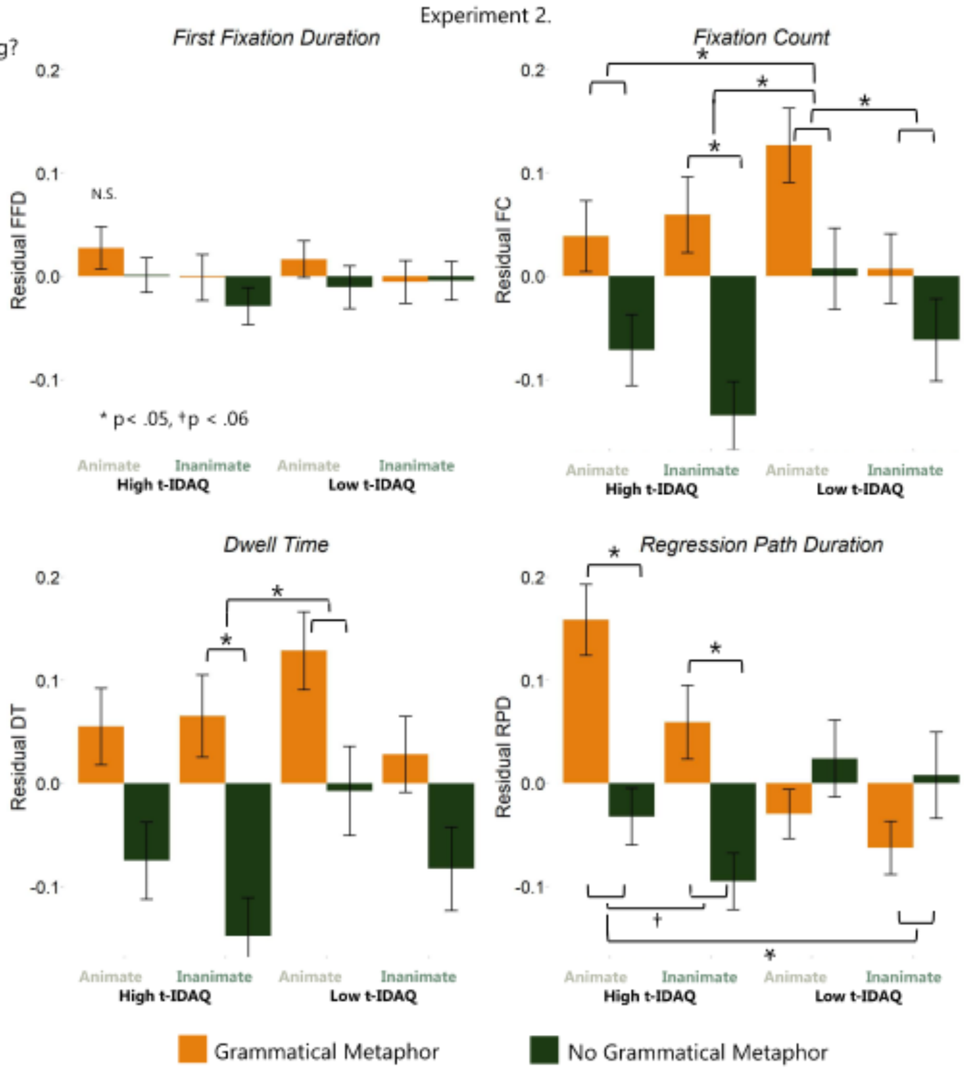
- 73 Participants recruited through Prolific
- Randomly assigned to **GM** / **no GM** framing
- Saw 18 items in each explanation condition
- Rated how much they agreed with each (animate/inanimate) explanation they saw

Experiment 2 (Eye-Tracking)

- 62 Participants recruited from UofSC
- Randomly assigned to **GM** / **no GM** framing
- Saw 18 items in each explanation condition
- Eyelink 1000 used to determine reading times on explanation verbs (e.g., judged / decoded)

EYE-TRACKING RESULTS

Experiment 2.



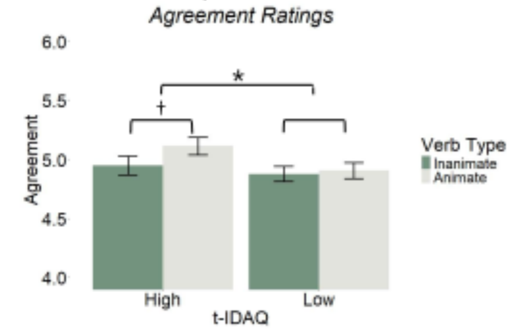
ANALYSIS

Experiment 1
- Ratings modeled using ordinal regression.

Experiment 2
- Data cleaned manually to correct vertical tracking errors
- First Fixation Duration, Fixation Count, Dwell Time, and Regression Path Duration on critical verbs were calculated as DVs
- All DVs were residualized using verb frequency, concreteness, and length (in letters)
- Residuals were modeled with Participants, Items, and Trial Order as random factors

BEHAVIORAL RESULTS

Experiment 1.



DISCUSSION

- The critical interaction of **Grammatical Metaphor** and **Explanation Verb Animacy**
  - Was **not** significant for the outcome measure in Ex 1. (Ratings) or the early processing measure in Ex 2. (FFD)
  - But **was** significant for middle/late processing measures in Ex. 2. (FC, DT, RPD)
- This interaction depended on participants' anthropomorphic beliefs about technology (t-IDAQ)
  - High t-IDAQ participants showed the predicted compatibility effect for **inanimate** verbs, which was reduced for **animate** verbs
  - Low t-IDAQ participants showed a cumulative inhibitory effects from **GM** and **Animacy**
- The latest process measure (RPD) patterned differently, as high t-IDAQ participants spent the longest time rereading the preceding text in the **GM** + **Animate** condition

DIGITAL POSTER & REFERENCES

References

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